

Statement of Purpose of Hrishikesh Belagali

My research interests are driven by a single question: how can we model complex physical systems effectively? The pursuit of this question has guided me across multiple computational frameworks, including molecular simulation, quantum algorithms, and machine learning techniques. By participating in the Science Undergraduate Laboratory Internship, I aim to deepen my understanding of quantum simulation and optimization methods by learning from experts in the field. This program aligns with my long-term goal of pursuing graduate studies in quantum algorithms and contributing to the development of novel computational methods for simulating complex systems, with applications in materials science and drug discovery.

As an undergraduate research assistant in the Vermaas Lab, I work on molecular dynamics simulations of biomolecular systems using VMD and NAMD. My tasks include setting up simulation systems, running simulations on high-performance computing clusters using SLURM, and analyzing the results using Tcl and Python scripts. I study the effect of small molecules such as butanol and isobutanol on bacterial membrane fluidity and stability. These simulations explain why such molecules disturb membranes and provide potential solutions to engineer more robust membranes, insights that are essential for industrial-scale biofuel production and difficult to obtain experimentally. This experience has shown me the power of computational approaches in understanding complex mechanisms at the molecular level.

At the same time, my experience highlighted the limitations of classical simulation. Simulating larger systems, capturing statistically rare events, and reaching biologically significant timescales are computationally expensive, which limits the scope of phenomena we can study. Quantum computing combines my passion for physics with computer science, and offers an alternative approach to modelling complex systems, potentially overcoming some of the challenges of classical methods. In order to explore my interest in quantum algorithms, I work as an Honors Research Scholar under Professor Ryan LaRose with a focus on generalized quantum simulators. My research involves studying group-theoretic methods and compilation optimization techniques in Python and Cirq to develop methods for simulating quantum algorithms on classical computers, which is essential for benchmarking and refining larger-scale quantum algorithms.

Alongside formal research, I have built several computational physics projects using Python, NumPy, Matplotlib, and Qiskit. I implemented a simulated annealing algorithm for minimizing Quadratic Unconstrained Binary Optimization (QUBO) problems, using stochastic tunneling to escape local minima and enhance convergence to the global minimum. I was curious to see whether quantum optimization algorithms would be able to achieve the same, so I formulated the same QUBO problems as Ising Hamiltonians and solved them using adiabatic evolution on IBM's Aer simulator. I faced challenges in maintaining the adiabatic condition throughout the simulation, which I addressed by trying different evolution schedules and adjusting total evolution time. I also had the opportunity to work on aqua-blue, an open-source machine learning package for predicting the time evolution of dynamical systems. I contributed by implementing time-lag and benchmarking common dynamical systems such as the Lorenz and Mackey-Glass systems. These projects have helped me develop strong Python programming skills, gain experience with scientific computing packages, and understand how to work with CI/CD workflows and version control software like Git and GitHub.

My combined experience in molecular dynamics, quantum algorithms and computational physics has cultivated my passion for contributing to research in methods for modelling physical mechanisms more efficiently. The SULI program at Oak Ridge National Laboratory is an important step towards this goal. ORNL's focus on biophysics simulations, their quantum-HPC initiative and research on hybrid quantum-classical algorithms closely align with my interests and are areas where my knowledge of molecular dynamics, scientific programming and quantum algorithms are directly applicable. Addi-

tionally, my familiarity with high-performance computing environments such as MSU ICER's HPCC, NCSA Delta and NCSA DeltaAI will allow me to quickly adapt to ORNL's computing resources and effectively contribute to ongoing projects. ORNL's interdisciplinary research environment will allow me to learn how to conduct research at a national laboratory, which will be beneficial for my career. I am eager to contribute my technical skills and passion for research to the program while expanding my knowledge of modelling techniques, which will provide invaluable preparation for graduate school and a career in research.